

AI@50, formally known as the "Dartmouth Artificial Intelligence Conference: The Next Fifty Years" (July 13–15, 2006), was a conference organized by James H. Moor, commemorating the 50th anniversary of the Dartmouth workshop which effectively inaugurated the history of artificial intelligence. Five of the original ten attendees were present: Marvin Minsky, Ray Solomonoff, Oliver Selfridge, Trenchard More, and John McCarthy.

While sponsored by Dartmouth College, General Electric, and the Frederick Whittemore Foundation, a \$200,000 grant to George Cybenko from the Defense Advanced Research Projects Agency (DARPA) called for a report of the proceedings that would:

Analyze progress on AI's original challenges during the first 50 years, and assess whether the challenges were "easier" or "harder" than originally thought and why

Document what the AI@50 participants believe are the major research and development challenges facing this field over the next 50 years, and identify what breakthroughs will be needed to meet those challenges

Relate those challenges and breakthroughs against developments and trends in other areas such as control theory, signal processing, information theory, statistics, and optimization theory.

A summary report by the conference director, James H. Moor, was published in AI Magazine.

Conference Program

(videos of the talks are available online)

James H. Moor, conference Director, Introduction

Carol Folt and Barry Scherr, Welcome

Carey Heckman, Tony Pandey and the Origins of Science

AI: Past, Present, Future

John McCarthy, What Was Expected, What We Did, and AI Today

Marvin Minsky, The Emotion Machine

The Future Model of Thinking

Ron Brachman and Hector Levesque, A Large Part of Human Thought

David Mumford, What is the Right Model for 'Thought'?

Stuart Russell, The Approach of Modern AI

The Future of Network Models

Geoffrey Hinton & Simon Osindero, From Pandemonium to Graphical Models and Back Again

Rick Granger, From Brain Circuits to Mind Manufacture

The Future of Learning & Search

Oliver Selfridge, Learning and Education for Software: New Approaches in Machine Learning

Ray Solomonoff, Machine Learning — Past and Future

Leslie Pack Kaelbling, Learning to be Intelligent

Peter Norvig, Web Search as a Product of and Catalyst for AI

The Future of AI

Rod Brooks, Intelligence and Bodies

Nils Nilsson, Routes to the Summit

Eric Horvitz, In Pursuit of Artificial Intelligence: Reflections on Challenges and Trajectories

The Future of Vision

Eric Grimson, Intelligent Medical Image Analysis: Computer Assisted Surgery and Disease

Monitoring

Takeo Kanade, Artificial Intelligence Vision: Progress and Non-Progress

Terry Sejnowski, A Critique of Pure Vision

The Future of Reasoning

Alan Bundy, Constructing, Selecting and Repairing Representations of Knowledge

Edwina Rissland, The Exquisite Centrality of Examples

Bart Selman, The Challenge and Promise of Automated Reasoning

The Future of Language and Cognition

Trenchard More The Birth of Array Theory and Nial

Eugene Charniak, Why Natural Language Processing is Now Statistical Natural Language Processing

Pat Langley, Intelligent Behavior in Humans and Machines

The Future of the Future

Ray Kurzweil, Why We Can Be Confident of Turing Test Capability Within a Quarter Century

George Cybenko, The Future Trajectory of AI

Charles J. Holland, DARPA's Perspective

AI and Games

Jonathan Schaeffer, Games as a Test-bed for Artificial Intelligence Research

Danny Kopec, Chess and AI

Shay Bushinsky, Principle Positions in Deep Junior's Development

Future Interactions with Intelligent Machines

Daniela Rus, Making Bodies Smart

Sherry Turkle, From Building Intelligences to Nurturing Sensibilities

Selected Submitted Papers: Future Strategies for AI

J. Storrs Hall, Self-improving AI: An Analysis

Selmer Bringsjord, The Logician Manifesto

Vincent C. Müller, Is There a Future for AI Without Representation?

Kristinn R. Thórisson, Integrated A.I. Systems

Selected Submitted Papers: Future Possibilities for AI

Eric Steinhart, Survival as a Digital Ghost

Colin T. A. Schmidt, Did You Leave That 'Contraption' Alone With Your Little Sister?

Michael Anderson & Susan Leigh Anderson, The Status of Machine Ethics

Marcello Guarini, Computation, Coherence, and Ethical Reasoning

References

External links

Dartmouth Artificial Intelligence Conference: The Next Fifty Years. Official conference Web site.

James Moor. The Dartmouth College Artificial Intelligence Conference: The Next Fifty Years. AI Magazine 27:4 [2006]: 87–91. ISSN 0738-4602. Official conference report, with photos; freely available online PDF.

Peter Norvig, Pictures from AI@50. Photographs of conference presenters.